DYLAN J LEE

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Technical Portfolio:
https://sites.google.com/view/dylan-lee-portfolio/home

SUMMARY

Mechanical engineering BS/MS student with experience designing and fabricating robotic mechanisms, precision positioning systems, and electromechanical prototypes. Former mechanical lead for UC San Diego's Yonder Dynamics rover team.

EDUCATION

University of California, San Diego
BS/MS Program, Mechanical Engineering BS June 2026 | MS Expected June 2027

- GPA: 3.73/4.00
- Coursework: Heat Transfer, Finite Element Methods, Solid/Fluid Mechanics, Signals, Controls, Senior Design

EXPERIENCE

YONDER DYNAMICS **Sep 2023–Jun 2026**
Mechanical Lead (Mar 2024–Jun 2026) | Chassis Project Manager (Apr 2025–Jun 2026)

- University Rover Challenge Mars rover project team at UCSD
- Designed, manufactured, and integrated a five-DOF robotic arm with over 4 ft of reach and sufficient mechanical precision for autonomous keyboard interaction.
- Reduced backlash **45-fold** (0.75° to 1 arcmin) at affected robotic-arm joints by replacing planetary gearboxes with harmonic drives.
- Led and managed a **25+** student mechanical team, leading training in SolidWorks, ANSYS, and machine shop practices (milling, laser cutting) to enable project success.

MULTI-CHAMBER-CAMERA BIOREACTOR (MCCB) — Senior Design Capstone **Jan 2026–Jun 2026**
Mechanical & Systems Design | Sponsored by Dr. Peter Chen, UC San Diego | Team of 4

- Co-developed a four-well platform that applies programmable electric (up to 1.5 V/cm) and magnetic (up to 5 mT, 250 Hz) fields to stem cells under a microscope, run from a Raspberry Pi touchscreen.
- Designed and fabricated the chassis: a 3030 T-slot frame with a 3/8 in acetal mounting plate (chosen over aluminum and steel to avoid eddy currents near the coils), motor mounts, and microscope platform.
- When the custom driver PCBs failed, routed and hand-soldered perfboard H-bridge modules for the electrode and electromagnet channels; the final integrated system ran on these boards.
- Wrote the ESP32 firmware that synthesizes sine, square, triangle, and step waveforms from 20 kHz PWM and an analog low-pass filter (coils to 250 Hz, electrodes to 100 Hz), with Hall-sensor readback at 2 kHz.
- Chose titanium electrodes over platinum and gold via a weighted scoring matrix (>2,000× cheaper per gram than Pt); validated the electrode drive to ±0.05 V/cm with an INA219 sensor and a saline-equivalent load.

ASTROLABE — Three-Axis Trackball and CAD Navigation Controller **06/2024–Present**
Independent Project | Sole Designer and Builder

- Independently designed and built a working trackball prototype, integrating custom mechanical hardware, electronics, and embedded control for desktop pointing and three-axis CAD navigation.
- Integrated two PMW3610 optical sensors and an nRF52840 microcontroller to reconstruct three-axis ball rotation, with Bluetooth and USB connectivity and ~30 hours of battery life.
- Used AI-assisted development to build embedded firmware and a Python Windows application with configurable controls and integrations for 10 CAD and 3D applications.
- Refined the design into a pocket-sized device that prints without supports and assembles without significant post-processing, at ~$30 per unit for small-scale production.
- Prototyped a 6-axis absolute object tracker using OpenCV AprilTag vision for 1:1 real-time object manipulation in Autodesk Fusion, with a foldable PLA/TPU tracked object made on toolchanger printers.

SKILLS & ACTIVITIES

- **Design & Analysis:** SolidWorks, Onshape, Autodesk Fusion, ANSYS Mechanical, GD&T
- **Manufacturing:** CNC and manual milling, additive manufacturing, laser cutting
- **Electronics:** KiCad, microcontrollers, motor drivers, perfboard prototyping
- **Programming:** MATLAB, Python, C++ (ESP32)
- Developed a curriculum for a non-profit youth engineering camp, designing activities and lessons to teach engineering principles and scientific concepts
- President of Redeemer's Grace Church's on-campus organization at UCSD
- Former president of FRC Team 5124, West Torrance Robotics